UCONN Mantis Cluster

This section explains how to access and work on the UConn Mantis HPC cluster.

Use this guide for:

- setting up your UConn/CAM account
- creating an SSH key
- connecting to Mantis
- using Visual Studio Code with Mantis
- installing tools and creating Conda environments
- finding the group working space
- submitting and checking SLURM jobs

Skip to our directory structure here.

Getting started

Before you can connect to the Mantis cluster, you need:

1. your **UConn NetID**
2. a **CAM user account**
3. an **SSH key**
4. access to the correct HPC project/account

Our UConn contact will provide your login details and UCONN NETID.

You can then create a CAM user account (enroll) and reset passwords here:

Enroll or reset CAM password

Instructions for setting up an SSH key are available here:

Setting up SSH keys

Connecting to Mantis

Once your CAM account and SSH key are set up, connect to Mantis from the command line using SSH:

```
ssh <user>@login.hpc.cam.uchc.edu
```

Replace `<user>` with your CAM username.

The full UConn getting started guide is available here:

UConn HPC getting started guide

This includes information on connecting to the cluster, running jobs, available software, and storage.

Connecting Visual Studio Code

Visual Studio Code, usually called **VS Code**, is a code editor (IDE) that makes it much easier to write, edit, search, and manage scripts. I like using VS Code because it lets me work with files on the cluster in a more visual and organised way, without needing to edit everything directly in the terminal using vim or nano.

A much easier workflow is to use VS Code with the **Remote - SSH** extension. This allows you to edit files on the cluster directly from your own computer, while the files themselves stay on the HPC.

Install VS Code and Remote SSH

First, install Visual Studio Code if you do not already have it, available here:

Download VS Code

Then, in VS Code, install the extension by searching `Remote – SSH`.

Add Mantis as an SSH host

In VS Code:

1. Press `Cmd + Shift + P`
2. Type:

```
Remote-SSH: Connect to Host
```

3. Choose:

```
Add New SSH Host
```

4. Enter your cluster SSH command, for example:

```
ssh your_username@login.hpc.cam.uchc.edu
```

5. Choose your SSH config file. This is usually:

```
~/.ssh/config
```

6. You can make reconnecting easier by adding a short host name to your SSH config file:

```
Host mantis
    HostName login.hpc.cam.uchc.edu
    User your_username
    IdentityFile ~/.ssh/id_rsa
```

Replace `your_username` with your CAM username.

The `IdentityFile` line should point to the private SSH key that you use to connect to Mantis. If your key has a different name, change this path.

Connect to Mantis from VS Code

After the SSH host has been added:

1. Press `Cmd + Shift + P`
2. Select:

```
Remote-SSH: Connect to Host
```

3. Choose:

```
login.hpc.cam.uchc.edu
```

Once connected, open your project folder on the cluster:

File → Open Folder

Then navigate to your project directory on Mantis.

You can now create and edit files normally in the VS Code editor, while the files remain stored on the cluster.

Why this is useful

Using VS Code with Remote SSH allows you to:

- open multiple files in tabs
- use syntax highlighting
- search across files and folders
- use the terminal inside VS Code on the cluster
- work with Git more easily
- avoid editing directly in vim most of the time

Downloading Tools and Creating Conda Environments

The Mantis cluster has many commonly used bioinformatics and computational tools available as modules.

Before installing a tool yourself, first check whether it is already available on the cluster:

```
module avail <tool>
```

Replace `<tool>` with the name of the software you are looking for.

For example: `module avail bcftools`

If the tool is available as a module, you can load it using:

```
module load <module_name>
```

If the tool is not available as a module, or if you need a specific version, you can create a **Conda environment**.

When creating or installing software into a Conda environment, use an **interactive job** rather than doing this directly on the login node. This helps prevent the login node from being slowed down or overwhelmed.

Instructions for setting up and using Conda environments on Mantis can be found here:

Conda environment set up

Group Working Space

Each user on the Mantis cluster has their own home directory. The home directory has approximately **1 TB of storage** and can only be accessed by the user. This is useful for scripts, working files, Conda environments, and user-specific configuration files.

We also have a **shared group working space** for final analysis outputs or files that other lab members can access. This is for the Stellenbosch Forest Genomics Team only.

The Stellenbosch group working directory is located at:

```
/core/labs/Wegrzyn/EucData/
```

We also have a shared working directory with collaborators at the **University of Pretoria**. The imputed data and 2000 gVCF files can be found in this directory.

The shared University of Pretoria directory is located at:

```
/core/projects/EBP/CBC/
```